Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Security+ Exam Prep (SY0-701)

Course Code: C628

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	22 July 2026	Initial release of the C628 courseware — 5 exam domains, 36 hands-on labs.	Dr. Alfred Ang
2.0	28 October 2026	Re-sequenced to the official SY0-701 blueprint (5 domains, 28 objectives); concept content expanded per objective; 52 labs aligned one-to-one with objectives (9 new labs); exam tips and practice questions per domain; 5-day schedule (one domain per day, 37.5 instructional hours) to match the published course duration.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Security+ Exam Prep (SY0-701)
Course Reference	C628
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 8 training hours per day (40 hours), of which 37.5 hours are instructional and the balance is tea breaks
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on practical labs
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain general security concepts — control types, CIA and AAA, Zero Trust, physical security, deception, change management and cryptography with PKI.
- LO2: Analyse threat actors, vectors and social engineering, application and platform vulnerabilities, indicators of attack, and mitigation techniques.
- LO3: Apply secure architecture — cloud and on-premises models, secure enterprise infrastructure, data protection, and resilience and recovery.
- LO4: Perform security operations — hardening, asset and vulnerability management, monitoring, security capabilities, IAM, automation and incident response.
- LO5: Support program oversight — governance, risk and third-party risk, compliance, audits and penetration testing, and security awareness.

Learning Reinforcement

- Every SY0-701 objective is taught first and then practised in at least one hands-on lab in the browser-based Killercoda playground.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress, plus a challenge task for fast finishers.
- Each objective closes with exam tips that highlight commonly confused terms and high-yield facts.
- Each domain closes with exam-style practice questions, reviewed together in the recap session.
- The trainer circulates during lab time to give individual feedback and unblock learners.

Course Schedule

Day 1 — Domain 1 — General Security Concepts (12%)

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, the SY0-701 exam format and domains, and

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		lab environment setup
10:00–11:00	60 min	**Domain 1 General Security Concepts — Objective 1.1 security controls and 1.2 fundamental concepts (CIA, AAA, Zero Trust, physical security, deception)**
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 1: Security Controls Categorization; Lab 2: CIA Triad and AAA on Linux; Lab 3: Deception: Honeypots, Honeyfiles and Honeytokens
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	**Objective 1.3 change management and 1.4 cryptographic solutions (encryption, hashing, signatures, PKI, key protection, obfuscation)**
15:00–16:30	90 min	Hands-on: Lab 4: Change Management with Version-Controlled Configuration; Lab 5: Hashing, Salting, and Digital Signatures
16:30–16:45	15 min	Tea break
16:45–17:45	60 min	Hands-on: Lab 6: Symmetric vs. Asymmetric Encryption and Key Exchange; Lab 7: PKI: Build Your Own Certificate Authority
17:45–18:30	45 min	Domain 1 recap, exam-style practice questions and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Domain 2 — Threats, Vulnerabilities and Mitigations (22%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 review and warm-up questions
9:45–10:30	45 min	**Domain 2 Threats, Vulnerabilities and Mitigations — Objective 2.1 threat actors and motivations, 2.2 threat vectors, attack surfaces and social engineering**
10:30–11:15	45 min	Hands-on: Lab 8: Threat Actors & the MITRE ATT&CK Framework; Lab 9: Phishing Email Analysis (SPF / DKIM / DMARC)
11:15–11:30	15 min	Tea break
11:30–12:00	30 min	**Objective 2.3 types of vulnerabilities (application, memory, web, OS, hardware, cloud, supply chain, zero-day)**
12:00–13:00	60 min	Hands-on: Lab 10: SQL Injection (SQLi); Lab 11: Cross-Site Scripting (XSS); Lab 12: Memory Vulnerabilities & Exploit Mitigations
13:00–14:00	60 min	Lunch break
14:00–14:45	45 min	**Objective 2.4 indicators of malicious activity (malware, physical, network, application, cryptographic and password attacks)**
14:45–16:15	90 min	Hands-on: Lab 13: Password Attacks: Spraying & Brute Force; Lab 14: Malware Static Analysis & Linux IOC Hunting; Lab 15: DoS Simulation and Detection
16:15–16:30	15 min	Tea break
16:30–16:50	20 min	**Objective 2.5 mitigation techniques used to secure the enterprise**
16:50–17:45	55 min	Hands-on: Lab 16: Linux Hardening Baseline with Lynis
17:45–18:30	45 min	Domain 2 recap, exam-style practice questions and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 3 — Domain 3 — Security Architecture (18%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 review and warm-up questions
9:45–10:20	35 min	**Domain 3 Security Architecture — Objective 3.1 architecture models (cloud, IaC, serverless, containers, IoT, ICS/SCADA)**
10:20–11:15	55 min	Hands-on: Lab 17: Cloud Shared Responsibility and IAM Least Privilege; Lab 18: Container Security with Docker
11:15–11:30	15 min	Tea break
11:30–12:10	40 min	**Objective 3.2 secure enterprise infrastructure (zones, appliances, port security, firewall types, VPN, SD-WAN, SASE)**
12:10–13:00	50 min	Hands-on: Lab 19: Network Segmentation and Security Zones; Lab 20:

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		Stateful Firewall with UFW and nftables
13:00–14:00	60 min	Lunch break
14:00–14:50	50 min	Hands-on: Lab 21: Web Application Firewall with ModSecurity and the OWASP CRS; Lab 22: Site-to-Site VPN with WireGuard
14:50–15:30	40 min	**Objective 3.3 data protection strategies and 3.4 resilience and recovery**
15:30–16:30	60 min	Hands-on: Lab 23: Data Classification and Data Loss Prevention; Lab 24: Data at Rest: LUKS2 Disk Encryption; Lab 25: Data in Transit: TLS Hardening and Cipher Audit
16:30–16:45	15 min	Tea break
16:45–17:45	60 min	Hands-on: Lab 26: Backup and Recovery with restic; Lab 27: Storage Resilience with Software RAID (mdadm)
17:45–18:30	45 min	Domain 3 recap, exam-style practice questions and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 4 — Domain 4 — Security Operations (28%)

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 review and warm-up questions
9:45–10:30	45 min	**Domain 4 Security Operations — Objectives 4.1 secure computing resources, 4.2 asset management, 4.3 vulnerability management**
10:30–11:30	60 min	Hands-on: Lab 28: Mobile and Endpoint Hardening Baseline; Lab 29: Application Security: Validation, SAST and Code Signing; Lab 30: Asset Inventory, Media Sanitisation and Disposal Certificate; Lab 31: Vulnerability Scanning, Confirmation and Rescan; Lab 32: Vulnerability Management Pipeline with Trivy
11:30–11:45	15 min	Tea break
11:45–12:15	30 min	**Objective 4.4 alerting and monitoring and 4.5 enterprise security capabilities**
12:15–13:00	45 min	Hands-on: Lab 33: SIEM Alerting and Monitoring with Wazuh; Lab 34: File Integrity Monitoring with AIDE
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	Hands-on: Lab 35: Secure Remote Access: SSH Hardening and Fail2ban; Lab 36: Network IDS with Suricata; Lab 37: Endpoint Detection and Response with osquery
15:00–15:40	40 min	**Objectives 4.6 identity and access management, 4.7 automation and orchestration, 4.8 incident response, 4.9 data sources for investigation**
15:40–16:30	50 min	Hands-on: Lab 38: Identity & Access Management with PAM and sudo; Lab 39: Multifactor Authentication with TOTP; Lab 40: Automation and Orchestration with a Mini-SOAR Playbook
16:30–16:45	15 min	Tea break
16:45–17:50	65 min	Hands-on: Lab 41: Incident Response: Live Triage; Lab 42: Digital Forensics: Acquisition & Chain of Custody; Lab 43: Packet Capture and Traffic Analysis with tcpdump and tshark; Lab 44: Log Investigation: Building an Attack Timeline
17:50–18:30	40 min	Domain 4 recap, exam-style practice questions and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 5 — Domain 5 — Security Program Management and Oversight (20%) & Exam Readiness

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 4 review and warm-up questions
9:45–10:30	45 min	**Domain 5 Security Program Management and Oversight — Objectives 5.1 governance, 5.2 risk management, 5.3 third-party risk**
10:30–11:30	60 min	Hands-on: Lab 45: Governance: Policies, Standards, Procedures; Lab 46: Risk

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		Register, CVSS Scoring, and Quantitative Risk; Lab 47: Supply Chain Security: SBOM and Vendor Trust
11:30–11:45	15 min	Tea break
11:45–12:20	35 min	**Objectives 5.4 compliance, 5.5 audits, assessments and penetration testing, 5.6 security awareness**
12:20–13:00	40 min	Hands-on: Lab 48: Compliance Audit with OpenSCAP
13:00–14:00	60 min	Lunch break
14:00–15:40	100 min	Hands-on: Lab 49: Passive Reconnaissance / OSINT; Lab 50: Active Reconnaissance with Nmap; Lab 51: Penetration Testing in an Authorised Lab; Lab 52: Security Awareness: Phishing Simulation
15:40–15:55	15 min	Tea break
15:55–17:00	65 min	Domain 5 recap and exam-style practice questions; full-course review across all 28 SY0-701 objectives
17:00–18:30	90 min	Exam readiness: timed practice-exam walkthrough, performance-based question strategy, booking the exam and next steps

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to the exam domains)

Exam domain	Exam weighting	Labs
Domain 1: General Security Concepts	12%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7
Domain 2: Threats, Vulnerabilities & Mitigations	22%	Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14, Lab 15, Lab 16
Domain 3: Security Architecture	18%	Lab 17, Lab 18, Lab 19, Lab 20, Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27
Domain 4: Security Operations	28%	Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36, Lab 37, Lab 38, Lab 39, Lab 40, Lab 41, Lab 42, Lab 43, Lab 44
Domain 5: Security Program Management & Oversight	20%	Lab 45, Lab 46, Lab 47, Lab 48, Lab 49, Lab 50, Lab 51, Lab 52

SY0-701 Exam Objective Coverage

The course follows the official CompTIA Security+ SY0-701 blueprint in order. Every exam objective is taught and then practised in at least one hands-on lab.

Objective	Exam objective (official wording)	Labs
1.1	Compare and contrast various types of security controls	Lab 1
1.2	Summarize fundamental security concepts	Lab 2, Lab 3
1.3	Explain the importance of change management processes and the impact to security	Lab 4
1.4	Explain the importance of using appropriate cryptographic solutions	Lab 5, Lab 6, Lab 7
2.1	Compare and contrast common threat actors and motivations	Lab 8
2.2	Explain common threat vectors and attack surfaces	Lab 9
2.3	Explain various types of vulnerabilities	Lab 10, Lab 11, Lab 12
2.4	Given a scenario, analyze indicators of malicious activity	Lab 13, Lab 14, Lab 15
2.5	Explain the purpose of mitigation techniques used to secure the enterprise	Lab 16
3.1	Compare and contrast security implications of different architecture models	Lab 17, Lab 18
3.2	Given a scenario, apply security principles to secure enterprise infrastructure	Lab 19, Lab 20, Lab 21, Lab 22

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3.3	Compare and contrast concepts and strategies to protect data	Lab 23, Lab 24, Lab 25
3.4	Explain the importance of resilience and recovery in security architecture	Lab 26, Lab 27
4.1	Given a scenario, apply common security techniques to computing resources	Lab 28, Lab 29
4.2	Explain the security implications of proper hardware, software, and data asset management	Lab 30
4.3	Explain various activities associated with vulnerability management	Lab 31, Lab 32
4.4	Explain security alerting and monitoring concepts and tools	Lab 33, Lab 34
4.5	Given a scenario, modify enterprise capabilities to enhance security	Lab 35, Lab 36, Lab 37
4.6	Given a scenario, implement and maintain identity and access management	Lab 38, Lab 39
4.7	Explain the importance of automation and orchestration related to secure operations	Lab 40
4.8	Explain appropriate incident response activities	Lab 41, Lab 42
4.9	Given a scenario, use data sources to support an investigation	Lab 43, Lab 44
5.1	Summarize elements of effective security governance	Lab 45
5.2	Explain elements of the risk management process	Lab 46
5.3	Explain the processes associated with third-party risk assessment and management	Lab 47
5.4	Summarize elements of effective security compliance	Lab 48
5.5	Explain types and purposes of audits and assessments	Lab 49, Lab 50, Lab 51
5.6	Given a scenario, implement security awareness practices	Lab 52

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